三、路由调试

1.启用所有设备的 ssh 服务，防火墙用户名 admin，明文密码Key-1122，其余设备用户名和明文密码均为 admin。

SW1、SW2、SW3和AC1：全局配置模式

```
ssh-server enable
!
```

RT1和RT2：全局配置模式

```
ip sshd enable
aaa authentication login default local //用本地用户信息进行认证
aaa authentication enable default none //禁用特权模式密码认证
!
```

FW1和FW2：全局配置模式

```
//在接口上配置ssh和ping
int 接口名称
manage ping
manage ssh
!
admin user admin   //进入管理员用户admin
password Key-1122  //配置admin的密码
: admin       //旧密码
```

2.配置所有设备的时区为GMT+08:00，调整 SW1 时间为实际时间，SW1配置为 ntp server，其他设备用 SW1 loopback1 ipv4 地址作为 ntp server 地址，ntp client 请求报文时间间隔 1 分钟。

sw1：全局配置模式

```
特权模式:
clock set 10:41:32 2023.08.02
show clock     //查询设备时间
全局配置模式:
ntp enable
ntp-service refclock-master 1  //设置参考时钟源进行同步
· 0表示该设备不作为参考时钟源，仅作为从其他设备同步时间。
· 1表示该设备是参考时钟源，其他设备将从该设备同步时间。
clock timezone  GMT add 8 0
ntp server 10.10.1.1    //看情况配置
ntp syn-interval 60
!
```

SW2、SW3和AC1：全局配置模式

```
ntp enable
ntp syn-interval 60
ntp server 10.10.1.1     //SW1的环回接口地址
clock timezone  GMT add 8 0     //配置时区
```

RT1和 RT2：全局配置模式

```
time-zone GMT 8 0
ntp query-interval 60
ntp server 10.10.1.1
!
```

FW1和FW2：全局配置模式

```
clock zone gmt
ntp  enable
ntp query-interval 60
ntp server  10.10.1.1
!
```

3.配置接口 ipv4 地址和 ipv6 地址，互联接口 ipv6 地址用本地链路地址。

```
关于IPv4v6地址和本地链路地址前面已经开启！！
```

4. SW2 配置 DHCPv4 和 DHCPv6，分别为总公司产品1（vlan10）段、总公司产品2段、分公司 Vlan130、分公司 Vlan140 和分公司 Vlan150 分配地址。IPv4地址池名称分别为 Poolv4-Vlan11、Poolv4-Vlan21、Poolv4-Vlan130、Poolv4-Vlan140、Poolv4-Vlan150，排除网关，DNS为10.10.210.101和 10.10.220.101。IPv6地 址池名称分别为Poolv6-Vlan11、Poolv6-Vlan21、Poolv6-Vlan130、Poolv6-Vlan140、Poolv6-Vlan150，IPv6 地址池用网络前缀表示,排除网关，DNS 为 2400:3200::1。PC1 保留地址 10.10.11.9 和 2001:10:1:11::9，PC2 保留地址10.10.21.9和 2001:10:1:21::9，AP1保留地址 10.10.130.9 和 2001:10:1:130::9。

SW1、AC1中继地址为 SW2 Loopback1 地址，SW1启用 DHCPv4 和DHCPv6 snooping，如果 E1/0/1 连接 dhcpv4 服务器，则关闭该端口，恢复时间为 10 分钟。

SW2：全局配置模式 IPV4

```
service dhcp
!
ip dhcp excluded-address 10.10.11.1
ip dhcp excluded-address 10.10.21.1
ip dhcp excluded-address 10.10.130.1
ip dhcp excluded-address 10.10.140.1
ip dhcp excluded-address 10.10.150.1
!
ip dhcp pool Poolv4-Vlan11
 network-address 10.10.11.0 255.255.255.0
 default-router 10.10.11.1
 dns-server 10.10.210.101 10.10.220.101
!
ip dhcp pool Poolv4-Vlan21
 network-address 10.10.21.0 255.255.255.0
 default-router 10.10.21.1
 dns-server 10.10.210.101 10.10.220.101
!
```

```
ip dhcp pool Poolv4-Vlan130
 network-address 10.10.130.0 255.255.255.0
 default-router 10.10.130.1
 dns-server 10.10.210.101 10.10.220.101
!
ip dhcp pool Poolv4-Vlan140
 network-address 10.10.140.0 255.255.255.0
 default-router 10.10.140.1
 dns-server 10.10.210.101 10.10.220.101
!
ip dhcp pool Poolv4-Vlan150
 network-address 10.10.150.0 255.255.255.0
 default-router 10.10.150.1
 dns-server 10.10.210.101 10.10.220.101
!
ip dhcp pool AP1
 host 10.10.130.9 255.255.255.0
 hardware-address (AP的mac地址)//WLAN-AP#get management detail   //读取AP地址信息
 default-router 10.10.130.1
 dns-server 10.10.210.101 10.10.220.101
!
ip dhcp pool PC1
 host 10.10.11.9 255.255.255.0
 hardware-address (PC机mac地址)
 default-router 10.10.11.1
 dns-server 10.10.210.101 10.10.220.101
```

```
!
ip dhcp pool PC2
 host 10.10.21.9 255.0.0.0
 hardware-address (PC机mac地址)
 default-router 10.10.21.1
 dns-server 10.10.210.101 10.10.220.101
!
```

SW2：全局配置模式IPV6

```
service dhcpv6
!
ipv6 dhcp pool Poolv6-Vlan150
 network-address 2001:10:1:150:: 64
 excluded-address 2001:10:1:150::1
 dns-server 2400:3200::1
!
ipv6 dhcp pool Poolv6-Vlan140
 network-address 2001:10:1:140:: 64
 excluded-address 2001:10:1:140::1
 dns-server 2400:3200::1
!
ipv6 dhcp pool Poolv6-Vlan130
 network-address 2001:10:1:130:: 64
static-binding 2001:10:1:130::9 (AP的mac地址)
 excluded-address 2001:10:1:130::1
 dns-server 2400:3200::1
!
ipv6 dhcp pool Poolv6-Vlan21
```

```
 network-address 2001:10:1:21:: 64
 static-binding 2001:10:1:21::9 (PC机mac地址)
 excluded-address 2001:10:1:21::1
 dns-server 2400:3200::1
!
ipv6 dhcp pool Poolv6-Vlan11
 network-address 2001:10:1:11:: 64
 static-binding 2001:10:1:11::9 (PC机mac地址)
 excluded-address 2001:10:1:11::1
 dns-server 2400:3200::1
!
interface Loopback1
 ipv6 dhcp server Poolv6-Vlan11
 ipv6 dhcp server Poolv6-Vlan130
 ipv6 dhcp server Poolv6-Vlan140
 ipv6 dhcp server Poolv6-Vlan150
!
//DHCPv6 Relay的目地地址取决于DHCPv6 Server将DHCPv6的地址池绑定到了那个三层接口上
interface Vlan10
 no ipv6 nd suppress-ra  //开启IPv6路由通告功能
 ipv6 nd managed-config-flag //地址配置时使用有状态DHCPv6协议
 ipv6 nd other-config-flag //通过HDCP方式获取其他信息
 ipv6 dhcp server Poolv6-Vlan21
!
```

SW1：全局配置模式

```
service dhcp
!
ip forward-protocol udp bootps //配置DHCP中继转发端口的UDP广播报文
!
ip dhcp snooping enable
!
service dhcpv6
!
savi enable
!
Interface Ethernet1/0/1
 ip dhcp snooping action shutdown recovery 600
!
Interface Ethernet1/0/22
 ip dhcp snooping trust
 ipv6 dhcp snooping trust
!
interface Vlan10
 no ipv6 nd suppress-ra
 ipv6 nd managed-config-flag
 ipv6 nd other-config-flag
 ip helper-address 10.10.10.1
 ipv6 dhcp relay destination 2001:10:1:2::1
!
```

AC1：全局配置模式

```
service dhcp
!
ip forward-protocol udp bootps
!
service dhcpv6
!
interface Vlan130
 no ipv6 nd suppress-ra
 ipv6 nd managed-config-flag
 ipv6 nd other-config-flag
 ip helper-address 10.10.10.1
 ipv6 dhcp relay destination 2001:10:1:2::1
!
interface Vlan140
 no ipv6 nd suppress-ra
 ipv6 nd managed-config-flag
 ipv6 nd other-config-flag
 ip helper-address 10.10.10.1
 ipv6 dhcp relay destination 2001:10:1:2::1
!
interface Vlan150
 no ipv6 nd suppress-ra
 ipv6 nd managed-config-flag
 ipv6 nd other-config-flag
 ip helper-address 10.10.10.1
```

```
 ipv6 dhcp relay destination 2001:10:1:2::1
!
```

5.SW1、SW2、SW3、RT1 以太链路、RT2 以太链路、FW1、FW2、AC1之间运行 OSPFv2 和 OSPFv3 协议（路由模式发布网络用接口地址，BGP协议除外）。 （1）SW1、SW2、SW3、RT1、RT2、FW1 之间 OSPFv2 和 OSPFv3 协议，进程 1，区域 0，分别发布 loopback1 地址路由和产品路由，FW1 通告type1 默认路由。

SW1：全局配置模式

```
router ospf 1
 ospf router-id 10.10.1.1
 network 10.10.1.1/32 area 0
 network 10.10.11.1/32 area 0
 network 10.10.255.1/32 area 0
 network 10.10.255.5/32 area 0
 network 10.10.255.14/32 area 0
!
```

```
router ipv6 ospf 1
 router-id 10.10.1.1
 passive-interface vlan 10
!
interface Vlan10
 ipv6 router ospf area 0 tag 1
!
interface Vlan1019
 ipv6 router ospf area 0 tag 1
!
interface Vlan1020
 ipv6 router ospf area 0 tag 1
!
interface Vlan1022
 ipv6 router ospf area 0 tag 1
!
interface Loopback1
 ipv6 router ospf area 0 tag 1
!
```

SW2：全局配置模式

```
router ospf 1
 ospf router-id 10.10.10.1
 network 10.10.10.1/32 area 0
 network 10.10.10.21/32 area 0
 network 10.10.255.2/32 area 0
 network 10.10.255.9/32 area 0
 network 10.10.255.22/32 area 0
!
```

```
router ipv6 ospf 1
 router-id 10.10.10.1
 passive-interface vlan 10
!
interface Vlan1019
 ipv6 router ospf area 0 tag 1
!
interface Vlan1020
 ipv6 router ospf area 0 tag 1
!
interface Vlan1022
 ipv6 router ospf area 0 tag 1
!
interface Vlan10
 ipv6 router ospf area 0 tag 1
!
interface Loopback1
 ipv6 router ospf area 0 tag 1
!
```

SW3：全局配置模式

```
router ospf 1
 ospf router-id 10.10.3.1
 network 10.10.3.1/32 area 0
 network 10.10.31.1/32 area 0
 network 10.10.255.6/32 area 0
 network 10.10.255.10/32 area 0
!
router ipv6 ospf 1
```

```
 router-id 10.10.3.1
 passive-interface vlan 10
!
interface Vlan1019
 ipv6 router ospf area 0 tag 1
!
interface Vlan1020
 ipv6 router ospf area 0 tag 1
!
interface Loopback1
 ipv6 router ospf area 0 tag 1
!
interface Vlan10
 ipv6 router ospf area 0 tag 1
!
```

RT1：全局配置模式

```
router ospf 1
 router-id 10.10.5.1
 network 10.10.5.1 255.255.255.255 area 0
 network 10.10.255.21 255.255.255.255 area 0
 network 10.10.255.18 255.255.255.255 area 0
 network 10.10.255.29 255.255.255.255 area 0
!
ipv6 unicast-routing
```

```
router ospfv3 1
 router-id 10.10.5.1
!
interface GigaEthernet0/0
 ipv6 ospf 1 area 0
!
interface GigaEthernet0/1
 ipv6 ospf 1 area 0
!
interface GigaEthernet0/2
 ipv6 ospf 1 area 0
!
interface Loopback1
 ipv6 ospf 1 area 0
!
```

RT2：全局配置模式

```
router ospf 1
 router-id 10.10.6.1
 network 10.10.6.1 255.255.255.255 area 0
 network 10.10.255.30 255.255.255.255 area 0
!
ipv6 unicast-routing
router ospfv3 1
 router-id 10.10.6.1
!
interface Loopback1
```

```
 ipv6 ospf 1 area 0
!
interface GigaEthernet0/0
 ipv6 ospf 1 area 0
!
```

FW1：全局配置模式

```
ip vrouter "trust-vr"
  router ospf 1
    router-id 10.10.7.1
    default-information originate always type 1
    network 10.10.7.1/32 area 0
    network 10.10.255.13/32 area 0
    network 10.10.255.17/32 area 0
  exit
  ipv6 router ospf 1
    router-id 10.10.7.1
    default-information originate always type 1
  exit
exit
interface ethernet0/1
  ipv6 ospf 1 area 0
exit
interface ethernet0/2
  ipv6 ospf 1 area 0
exit
```

配置验证：特权模式

```
show running-config router ospf   //（交换机）查看router ospf配置
show ip ospf neighbor             //查看ospf邻居
```

（2）RT2 与 AC1 之间运行 OSPFv2 协议，进程 1，nssa no-summary 区域 1；AC1发布loopback1地址路由、产品和营销路由，用prefix-lis重发布 loopback3。

RT2:全局配置模式

```
router ospf 1
 network 10.10.255.41 255.255.255.255 area 1
 area 1 nssa no-summary
!
```

AC1：全局配置模式

```
ip prefix-list l3 seq 5 permit 10.10.4.3/32
!
route-map l3 permit 10
 match ip address prefix-list l3
```

```
!
router ospf 1
 ospf router-id 10.10.4.1
 area 1 nssa no-summary
 network 10.10.4.1/32 area 1
 network 10.10.130.1/32 area 1
 network 10.10.140.1/32 area 1
 network 10.10.150.1/32 area 1
 network 10.10.255.42/32 area 1
 redistribute connected route-map l3
!
```

验证配置：特权模式

```
show running-config router ospf   //（交换机）查看router ospf配置
show ip ospf neighbor             //查看ospf邻居
```

（3）RT2与AC1之间运行OSPFv3协议，进程1，stub no-summary区域1；AC1发布loopback1地址路由、产品和营销。

AC1：全局配置模式

```
router ipv6 ospf 1
 router-id 10.10.4.1
```

```
 area 1 stub no-summary
!
interface Vlan140
 ipv6 router ospf area 1 tag 1
!
interface Vlan150
 ipv6 router ospf area 1 tag 1
!
interface Vlan1001
 ipv6 router ospf area 1 tag 1
!
interface Loopback1
 ipv6 router ospf area 1 tag 1
!
```

RT2：全局配置模式

```
interface GigaEthernet0/1
 ipv6 ospf 1 area 1
!
router ospfv3 1
 area 1 stub no-summary
!
```

配置验证：特权模式

```
show ip ospf neighbor                //查看ospf邻居
show ipv6 ospf neighbor
```

（4）SW3模拟办事处产品和营销接口配置为loopback，模拟接口up。SW3模拟办事处与 FW2 之间运行 OSPFv2 协议，进程 2，区域 2，SW3模拟办事处发布 loopback2、产品和营销。SW3模拟办事处配置ipv6默认路由；FW2分别配置到SW3模拟办事处 loopback2、产品和营销的ipv6明细静态路由，FW2重发布静态路由到OSPFv3协议。

SW3：全局配置模式

```
interface e1/0/11-12
loopback
!
router ospf 2 vrf Office
 ospf router-id 10.10.3.2
 network 10.10.3.2/32 area 2
 network 10.10.110.1/32 area 2
 network 10.10.120.1/32 area 2
 network 10.10.255.46/32 area 2
!
ipv6 route vrf Office ::/0 fe80::203:fff:fe2b:6061 Vlan1015
!                                   //（对端设备FW2e0/1接口的link-local）
```

FW 2：全局配置模式

```
ip vrouter "trust-vr"
  ipv6 route 2001:10:1:3::2/128 "ethernet0/1" FE80::203:FFF:FE27:FD70 //（对端设备SW3vlan1015的link-local）
  ipv6 route 2001:10:1:120::1/128 "ethernet0/1" FE80::203:FFF:FE27:FD70
  ipv6 route 2001:10:1:110::1/128 "ethernet0/1" FE80::203:FFF:FE27:FD70
  router ospf 2
    router-id 10.10.8.1
    network 10.10.255.45/32 area 0.0.0.2
    network 10.10.255.26/32 area 0.0.0.2
    network 10.10.8.1/32 area 0.0.0.2
```

```
  exit
  ipv6 router ospf 2
    router-id 10.10.8.1
    redistribute static
  exit
exit
```

配置验证：特权模式

```
show ipv6 route  //（防火墙）查看ipv6路由表
show configuration vrouter  //（防火墙）查看当前路由配置
show ipv6 route vrf Office   //（交换机）查看ipv6路由表
show interface ethernet status  //（交换机）查看接口状态
show running-config router ospf  //（交换机）查看当前ospf配置
```

（5）RT1、FW2之间OSPFv2和OSPFv3 协议，进程 2，区域 2；RT1 发布 loopback4 路由，向该区域通告 type1 默认路由；FW2 发布 loopback1 路由，FW2 禁止学习到集团和分公司的所有路由。RT1用 prefix-list匹配 FW2 loopback1路由、SW3 模拟办事处 loopback2和产品路由、RT1 与 FW2 直连 ipv4 路由，将这些路由重发布到区域 0。

RT1：全局配置模式

```
router ospf 2
 router-id 10.10.5.4
 network 10.10.5.4 255.255.255.255 area 2
 network 10.10.255.25 255.255.255.255 area 2
 default-information originate always metric-type 1
!
router ospfv3 2
 router-id 10.10.5.4
```

```
 default-information originate metric-type 1
!
interface GigaEthernet0/3
 ipv6 ospf 2 area 2
!
interface Loopback1
 ipv6 ospf 1 area 0
!
route-map C 10 permit
 match ip address prefix-list C
!
route-map Office 10 permit
 match ip address prefix-list Office
!
ip prefix-list C seq 5 permit 10.10.255.24/30
ip prefix-list Office seq 5 permit 10.10.8.1/32
ip prefix-list Office seq 10 permit 10.10.3.2/32
ip prefix-list Office seq 15 permit 10.10.110.0/24
!
router ospf 1
 redistribute connect route-map C
 redistribute ospf 2 route-map Office
!
```

FW2:全局配置模式

```
ip vrouter "trust-vr"
  router ospf 2
    network 10.10.255.26/32 area 0.0.0.2
  exit
exit
interface ethernet0/2
  ipv6 ospf 2 area 2
exit
interface loopback1
  ipv6 ospf 2 area 2
exit
```

配置验证：特权模式

```
show configuration vrouter  //（防火墙）查看当前路由配置
show configuration interface loopback1 //（防火墙）查看当前接口配置状态
show ip ospf neighbor  //查看ospf邻居
show running-config interface g0/3 //（路由器）查看当前接口配置
```

（6）修改 ospf cost 为 100，实现 SW1 分别与 RT2、FW2 之间 ipv4 和 ipv6 互访流量优先通过 SW1_SW2_RT1 链路转发，SW2 访问 Internet ipv4 和 ipv6 流量优先通过 SW2_SW1_FW1 链路转发。

FW1：全局配置模式

```
interface ethernet0/2
  ip ospf cost 100
  ipv6 ospf cost 100
exit
```

```
FW1# show ip ospf interface
ethernet0/1 is up, line protocol is up
  Internet Address 10.10.255.13/30, Area 0.0.0.0
  Process ID 1, Router ID 10.10.7.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 10.10.7.1, Interface Address 10.10.255.13
  Backup Designated Router (ID) 10.10.1.1, Interface Address 10.10.255.14
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:03
  Neighbor Count is 1, Adjacent neighbor count is 1
ethernet0/2 is up, line protocol is up
  Internet Address 10.10.255.17/30, Area 0.0.0.0
  Process ID 1, Router ID 10.10.7.1, Network Type BROADCAST, Cost: 100
  Transmit Delay is 1 sec, State BDR, Priority 1
  Designated Router (ID) 10.10.5.1, Interface Address 10.10.255.18
  Backup Designated Router (ID) 10.10.7.1, Interface Address 10.10.255.17
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:08
  Neighbor Count is 1, Adjacent neighbor count is 1
```

RT1：全局配置模式

```
interface GigaEthernet0/2
 ip ospf cost 100
 ipv6 ospf cost 100
 !
```

```
GigaEthernet0/2 is up, line protocol is UP
  Internet Address: 10.10.255.18/30
  Interface index: 4, 0x3587db50
  Nettype: Broadcast
  OSPF process is 1,  AREA: 0, Router ID: 10.10.5.1
  Cost: 100, Priority 1, Transmit Delay is 1 sec
  Hello interval is 10, Dead timer is 40, Retransmit is 5
  OSPF INTF State is IDr
  Designated Router ID: 10.10.5.1, Interface address 10.10.255.18
  Backup Designated Router ID: 10.10.7.1, Interface address 10.10.255.17
  Neighbor Count is 1, Adjacent neighbor count is 1
    Adjacent with neighbor 10.10.7.1 (Backup Designated Router)
  Null authentication enabled
```

配置验证：特权模式

```
show running-config interface g0/2 //（路由器）查看当前接口配置
show configuration interface e0/2 //（防火墙）查看当前接口配置
show ip ospf interface        //查看接口cost值
在 OSPF 中，cost 值越大，意味着通过该链路传输数据包的成本越高，反之，cost 值越小，则表示数据包的传输成本越低。
```

6.（2）RT1 的 S1/0 与 RT2 的 S1/1 之间采用 chap 双向认证，用户名为对端设备名称，密码为 Key-1122。

RT1：全局配置模式

```
aaa authentication ppp default local
username RT1 password 0 Key-1122
interface Serial1/0
 encapsulation ppp
 ppp authentication chap
 ppp chap hostname RT2
 ppp chap password 0 Key-1122
 !
 interface Serial1/1
 encapsulation ppp
!
```

RT2:全局配置模式

```
aaa authentication ppp default local
username RT2 password 0 Key-1122
interface Serial1/0
 encapsulation ppp
!
interface Serial1/1
 encapsulation ppp
 ppp authentication chap
 ppp chap hostname RT1
 ppp chap password 0 Key-1122
!
```

配置验证：特权模式

```
show running-config interface s1/0  //查看当前接口配置
```

6. RT1 串行链路、RT2 串行链路、FW1、AC1 之间分别运行 RIP和 RIPng 协议，FW1、RT1、RT2 的 RIP 和 RIPng 发布 loopback2 地址路由，AC1 RIP 发布 loopback2 地址路由，AC1 RIPng 采用 route-map 匹配prefix-list 重发布 loopback2 地址路由。RT1 配置 offset 值为 3 的路由策略，实现 RT1-S1/0_RT2-S1/1 为主链路，RT1-S1/1_RT2-S1/0 为备份链路，ipv4 的 ACL 名称为AclRIP，ipv6 的ACL 名称为 AclRIPng。

FW1：全局配置模式

```
ip vrouter "trust-vr"
  router rip
    network 10.10.255.16/30
    network 10.10.7.2/32
  exit
  ipv6 router rip
    network loopback2
    network ethernet0/2
  exit
exit
```

AC1：全局配置模式

```
ipv6 prefix-list L2 seq 5 permit 2001:10:1:4::2/128
route-map L2 permit 10
 match ipv6 address prefix-list L2
!
router ipv6 rip
 redistribute connected route-map L2
!
router rip
 network 10.10.4.2/32
 network 10.10.255.40/30
!
interface Vlan1001
 ipv6 router rip
!
```

RT2：全局配置模式

```
router rip 1
 version 2
 no auto-summary
!
router ripng 1
!
interface Serial1/0
 ip rip 1 enable
 ipv6 rip 1 enable
!
interface Serial1/1
 ip rip 1 enable
 ipv6 rip 1 enable
!
interface Loopback2
 ip rip 1 enable
 ipv6 rip 1 enable
!
interface GigaEthernet0/1
 ip rip 1 enable
 ipv6 rip 1 enable
!
```

RT1：全局配置模式

```
router rip 1
 version 2
 no auto-summary
```

```
!
router ripng 1
!
interface Serial1/0
 ip rip 1 enable
 ipv6 rip 1 enable
!
interface Serial1/1
 ip rip 1 enable
 ipv6 rip 1 enable
!
interface GigaEthernet0/2
 ip rip 1 enable
 ipv6 rip 1 enable
!
interface Loopback2
 ip rip 1 enable
 ipv6 rip 1 enable
!
ip access-list standard AclRIP
 permit any  sequence 10
!
ipv6 access-list AclRIPng
 permit ipv6 any any sequence 10
!
router rip 1
 offset Serial1/1 in AclRIP 3
 offset Serial1/1 out AclRIP 3
!
router ripng 1
```

```
 offset Serial1/1 in AclRIPng 3
 offset Serial1/1 out AclRIPng 3
!
```

配置验证：特权模式

```
show ip rip database //（交换机和防火墙）查看RIP所有路由
show ip rip 1 database //（路由器）查看RIP所有路由
在路由器中，偏移量（Offset）是IP数据报头中一个用于报文分段的字段，它用来描述IP分段的位置和拼接方式，以确保数据包成功的传输。
路由器默认的Offset值是0，这是由IP协议规定的。
Offset值越小，表示该数据包分片相对于原始数据包的位置越靠前，因此在重组时，优先重组 Offset值小的包，从而可以更快地组装成完整的原始数据包。
```

7.RT1 以太链路、RT2 以太链路之间运行 ISIS 协议，进程 1，分别实现 loopback3 之间 ipv4 互通和 ipv6 互通。RT1、RT2 的 NET 分别为 10.0000.0000.0001.00 、10.0000.0000.0002.00 ， 路 由 器 类 型 是 Level-2，接口网络类型为点到点。配置域 md5 认证和接口 md5 认证， 密码均为 Key-1122。

RT1：全局配置模式

```
router isis 1
 is-type level-2
 authentication mode md5
 authentication key 0 Key-1122
 net 10.0000.0000.0001.00
!
interface GigaEthernet0/0
 ip router isis 1
 ipv6 router isis 1
 isis network point-to-point
 isis authentication mode md5
 isis authentication key 0 Key-1122
!
interface Loopback3
```

```
 ip router isis 1
 ipv6 router isis 1
!
```

RT2：全局配置模式

```
router isis 1
 is-type level-2
 authentication mode md5
 authentication key 0 Key-1122
 net 10.0000.0000.0002.00
!
interface GigaEthernet0/0
 ip router isis 1
 ipv6 router isis 1
 isis network point-to-point
 isis authentication mode md5
 isis authentication key 0 Key-1122
!
interface Loopback3
 ip router isis 1
 ipv6 router isis 1
!
```

配置验证：特权模式

```
show isis route                    //查看isis路由表
show isis neighbors                //查看isis邻居
```

8.RT2 配置 ipv4 nat，实现 AC1 ipv4 产品用 RT2 外网接口 ipv4地址访问 Internet。RT2 配置 nat64，实现 AC1 ipv6 产品用 RT2 外网接口 ipv4 地址访问 Internet，ipv4 地址转 ipv6 地址前缀为64:ff9b::/96。

RT2：全局配置模式

```
ip access-list standard natv4
 permit 10.10.140.1 255.255.255.0 sequence 10
!
ipv6 access-list natv6
 permit ipv6 2001:10:1:140::/64 any sequence 10
!
ip nat inside source list natv4 interface GigaEthernet0/3
ipv6 nat v6v4 source list natv6 interface GigaEthernet0/3
ipv6 nat prefix 64:FF9B::/96 v4-mapped natv6
!
interface GigaEthernet0/1
 ip nat inside
 ipv6 nat
```

```
!
interface GigaEthernet0/3
 ip nat outside
 ipv6 nat
!
ip route default 200.200.200.5
```

```
AC1#ping src 10.10.140.1 200.200.200.5
Type ^c to abort.
Sending 5 56-byte ICMP Echos to 200.200.200.5, using source address 10.10.140.1, timeout is 2 seconds.
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/10/20 ms
AC1#
AC1#ping6 src 2001:10:1:140::1 64:ff9b::200.200.200.5

Type ^c to abort.
Sending 5 56-byte ICMP Echos to 64:ff9b::200.200.200.5, using src address 2001:10:1:140::1, timeout is 2 seconds.
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 0/8/10 ms
```

```
RT2#show ip nat translations
Pro. Dir Inside local       Inside global      Outside local      Outside global
ICMP OUT 10.10.140.1:4660   200.200.200.6:4660 200.200.200.5:4660 200.200.200.5:4660
RT2#show ipv6 nat translations
Prot  V6Src              V6Dst              V4Src              V4Dst
ICMP  2001:10:1:140::1-466064:FF9B::C8C8:C805-4660200.200.200.6-4660200.200.200.5-4660
```

配置验证：特权模式

```
show ip nat statistics      //查看nat地址转换状态
在AC1上执行ping6 src 2001:10:1:140::1 64:ff9b::200.200.200.5命令测试成功之后在RT2执行show ipv6 nat translation命令
```

9.SW1、SW2、SW3、RT1、RT2 之间运行 BGP 协议，SW1、SW2、RT1 AS号 65001、RT2 AS 号 65002、SW3 AS 号 65003。 （1）SW1、SW2、SW3、RT1、RT2 之间通过 loopback1 建立 ipv4 和 ipv6 BGP 邻居。SW1 和 SW2 之间财务通过 loopback2 建立 ipv4 BGP邻居，SW1 和 SW2 的 loopback2 互通采用静态路由。

SW1：全局配置模式

```
interface Loopback2
 ip vrf forwarding Finance
 ipv6 address 2001:10:1:1::2/128
 ip address 10.10.1.2 255.255.255.255
!
router bgp 65001
 bgp router-id 10.10.1.1
 neighbor 10.10.3.1 remote-as 65003
 neighbor 10.10.3.1 ebgp-multihop 255
 neighbor 10.10.3.1 update-source 10.10.1.1
 neighbor 10.10.5.1 remote-as 65001
```

```
 neighbor 10.10.5.1 update-source 10.10.1.1
 neighbor 10.10.10.1 remote-as 65001
 neighbor 10.10.10.1 next-hop-self
 neighbor 10.10.10.1 update-source 10.10.1.1
 neighbor 2001:10:1:3::1 remote-as 65003
 neighbor 2001:10:1:3::1 ebgp-multihop 255
 neighbor 2001:10:1:3::1 update-source 2001:10:1:1::1
 no neighbor 2001:10:1:3::1 activate
 neighbor 2001:10:1:5::1 remote-as 65001
 neighbor 2001:10:1:5::1 update-source 2001:10:1:1::1
 no neighbor 2001:10:1:5::1 activate
 neighbor 2001:10:1:2::1 remote-as 65001
 neighbor 2001:10:1:2::1 update-source 2001:10:1:1::1
 no neighbor 2001:10:1:2::1 activate
 address-family ipv6 unicast
 neighbor 2001:10:1:3::1 activate
 neighbor 2001:10:1:5::1 activate
 neighbor 2001:10:1:2::1 activate
neighbor 2001:10:1:2::1 next-hop-self
 exit-address-family
 address-family ipv4 vrf Finance
 neighbor 10.10.10.2 remote-as 65001
 neighbor 10.10.10.2 update-source 10.10.1.2
```

```
 exit-address-family
!
ip route vrf Finance 10.10.10.2/32 10.10.255.2
```

SW2:全局配置模式

```
interface Loopback2
 ip vrf forwarding Finance
 ipv6 address 2001:10:1:2::2/128
 ip address 10.10.10.2 255.255.255.255
!
router bgp 65001
 bgp router-id 10.10.10.1
 neighbor 10.10.1.1 remote-as 65001
 neighbor 10.10.1.1 next-hop-self
 neighbor 10.10.1.1 update-source 10.10.10.1
 neighbor 10.10.3.1 remote-as 65003
 neighbor 10.10.3.1 ebgp-multihop 255
 neighbor 10.10.3.1 update-source 10.10.10.1
 neighbor 10.10.5.1 remote-as 65001
 neighbor 10.10.5.1 update-source 10.10.10.1
 neighbor 10.10.5.1 next-hop-self
 neighbor 2001:10:1:1::1 remote-as 65001
 neighbor 2001:10:1:1::1 update-source 2001:10:1:2::1
 no neighbor 2001:10:1:1::1 activate
```

```
 neighbor 2001:10:1:3::1 remote-as 65003
 neighbor 2001:10:1:3::1 ebgp-multihop 255
 neighbor 2001:10:1:3::1 update-source 2001:10:1:2::1
 no neighbor 2001:10:1:3::1 activate
 neighbor 2001:10:1:5::1 remote-as 65001
 neighbor 2001:10:1:5::1 update-source 2001:10:1:2::1
 no neighbor 2001:10:1:5::1 activate
 address-family ipv6 unicast
 neighbor 2001:10:1:1::1 activate
 neighbor 2001:10:1:1::1 next-hop-self
 neighbor 2001:10:1:3::1 activate
 neighbor 2001:10:1:5::1 activate
 neighbor 2001:10:1:5::1 next-hop-self
 exit-address-family
 address-family ipv4 vrf Finance
 neighbor 10.10.1.1 remote-as 65001
 neighbor 10.10.1.1 update-source 10.10.10.2
 exit-address-family
!
ip route vrf Finance 10.10.1.2/32 10.10.255.1
```

SW3：全局配置模式

```
router bgp 65003
 bgp router-id 10.10.3.1
```

```
 neighbor 10.10.1.1 remote-as 65001
 neighbor 10.10.1.1 ebgp-multihop 255
 neighbor 10.10.1.1 update-source 10.10.3.1
 neighbor 10.10.10.1 remote-as 65001
 neighbor 10.10.10.1 ebgp-multihop 255
 neighbor 10.10.10.1 update-source 10.10.3.1
 neighbor 2001:10:1:1::1 remote-as 65001
 neighbor 2001:10:1:1::1 ebgp-multihop 255
 neighbor 2001:10:1:1::1 update-source 2001:10:1:3::1
 no neighbor 2001:10:1:1::1 activate
 neighbor 2001:10:1:2::1 remote-as 65001
 neighbor 2001:10:1:2::1 ebgp-multihop 255
 neighbor 2001:10:1:2::1 update-source 2001:10:1:3::1
 no neighbor 2001:10:1:2::1 activate
 address-family ipv6 unicast
 neighbor 2001:10:1:1::1 activate
 neighbor 2001:10:1:2::1 activate
 exit-address-family
!
```

RT1：全局配置模式

```
router bgp 65001
 bgp router-id 10.10.5.1
```

```
 neighbor 10.10.1.1 remote-as 65001
 neighbor 10.10.1.1 update-source Loopback1
 neighbor 10.10.6.1 remote-as 65002
 neighbor 10.10.6.1 ebgp-multihop 255
 neighbor 10.10.6.1 update-source Loopback1
 neighbor 10.10.10.1 remote-as 65001
 neighbor 10.10.10.1 update-source Loopback1
 neighbor 10.10.10.1 next-hop-self
 neighbor 2001:10:1:1::1 remote-as 65001
 neighbor 2001:10:1:1::1 update-source Loopback1
 neighbor 2001:10:1:6::1 remote-as 65002
 neighbor 2001:10:1:6::1 ebgp-multihop 255
 neighbor 2001:10:1:6::1 update-source Loopback1
 neighbor 2001:10:1:2::1 remote-as 65001
 neighbor 2001:10:1:2::1 update-source Loopback1
 address-family ipv4
 no neighbor 2001:10:1:1::1 activate
 no neighbor 2001:10:1:6::1 activate
 no neighbor 2001:10:1:2::1 activate
 exit-address-family
 address-family ipv6
 neighbor 2001:10:1:2::1 next-hop-self
 neighbor 2001:10:1:1::1 activate
```

```
 neighbor 2001:10:1:6::1 activate
 neighbor 2001:10:1:2::1 activate
 exit-address-family
!
```

RT2:全局配置模式

```
router bgp 65002
 bgp router-id 10.10.6.1
 neighbor 10.10.5.1 remote-as 65001
 neighbor 10.10.5.1 ebgp-multihop 255
 neighbor 10.10.5.1 update-source Loopback1
 neighbor 2001:10:1:5::1 remote-as 65001
 neighbor 2001:10:1:5::1 ebgp-multihop 255
 neighbor 2001:10:1:5::1 update-source Loopback1
 address-family ipv4
 no neighbor 2001:10:1:5::1 activate
 exit-address-family
 address-family ipv6
 neighbor 2001:10:1:5::1 activate
 exit-address-family
!
```

配置验证：特权模式

```
show ip bgp summary     //查看ipv4 bgp邻居
show ip bgp ipv6 unicast summary  //(路由器）查看ipv6 bgp邻居
```

（2）SW1、SW2、SW3、RT2 分别只发布营销、法务、财务、人力等 ipv4 和 ipv6 路由；RT1 发布办事处营销 ipv4 和 ipv6 路由到 BGP。

SW1:全局配置模式

```
router bgp 65001
 network 10.10.12.0/24
 network 10.10.13.0/24
 network 10.10.15.0/24
 address-family ipv6 unicast
 network 2001:10:1:12::/64
 network 2001:10:1:13::/64
 network 2001:10:1:15::/64
 exit-address-family
 address-family ipv4 vrf Finance
 network 10.10.14.0/24
 exit-address-family
!
```

SW2：全局配置模式

```
router bgp 65001
 network 10.10.22.0/24
 network 10.10.23.0/24
 network 10.10.25.0/24
 address-family ipv6 unicast
 network 2001:10:1:22::/64
 network 2001:10:1:23::/64
 network 2001:10:1:25::/64
 exit-address-family
 address-family ipv4 vrf Finance
 network 10.10.24.0/24
 exit-address-family
!
```

SW3：全局配置模式

```
router bgp 65003
 network 10.10.35.0/24
 network 10.10.32.0/24
 network 10.10.33.0/24
 address-family ipv6 unicast
 network 2001:10:1:32::/64
 network 2001:10:1:33::/64
 network 2001:10:1:35::/64
 exit-address-family
!
```

RT1：全局配

```
router bgp 65001
 network 10.10.120.0/24
 address-family ipv6
 network 2001:10:1:120::/64
 exit-address-family
!
```

RT2：全局配置模式

```
router bgp 65002
 network 10.10.150.0/24
 address-family ipv6
 network 2001:10:1:150::/64
 exit-address-family
!
```

配置验证：特权模式

```
show bgp ipv6     //（交换机）查看IPv6 bgp路由表
show ip bgp      //(交换机、路由器)查看IPv4 bgp路由器
show ip bgp ipv6 unicast  //（路由器）查看IPv6 bgp路由表
```

（3）SW3 营销分别与 SW1 和 SW2 营销 ipv4 和 ipv6 互访优先在 SW3_SW1 链路转发；SW3 法务及人力分别与 SW1 和 SW2 法务及人力 ipv4和 ipv6 互访优先在 SW3_SW2 链路转发，主备链路相互备份；用 prefix-list、route-map 和 BGP 路径属性进行选路，新增 AS 65000。

SW2：全局配置模式

```
ip prefix-list SW3v4 seq 5 permit 10.10.32.0/24
!
ipv6 prefix-list SW3v6 seq 5 permit 2001:10:1:32::1/64
!
route-map SW3v6 permit 10
 match ipv6 address prefix-list SW3v6
 set as-path prepend 65000
!
route-map SW3v6 permit 20
!
route-map SW3v4 permit 10
 match ip address prefix-list SW3v4
 set as-path prepend 65000
```

```
!
route-map SW3v4 permit 20
!
router bgp 65001
 neighbor 10.10.3.1 route-map SW3v4 in
 address-family ipv6 unicast
 neighbor 2001:10:1:3::1 route-map SW3v6 in
 exit-address-family
!
```

SW1：全局配置模式

```
ip prefix-list SW3v4 seq 5 permit 10.10.33.0/24
ip prefix-list SW3v4 seq 10 permit 10.10.35.0/24
!
ipv6 prefix-list SW3v6 seq 5 permit 2001:10:1:33::/64
ipv6 prefix-list SW3v6 seq 10 permit 2001:10:1:35::/64
!
route-map SW3v4 permit 10
 match ip address prefix-list SW3v4
 set as-path prepend 65000
!
route-map SW3v4 permit 20
!
route-map SW3v6 permit 10
```

```
 match ipv6 address prefix-list SW3v6
 set as-path prepend 65000
!
route-map SW3v6 permit 20
!
router bgp 65001
 neighbor 10.10.3.1 route-map SW3v4 in
 address-family ipv6 unicast
 neighbor 2001:10:1:3::1 route-map SW3v6 in
 exit-address-family
!
```

SW3：全局配置模式

```
ip prefix-list SW1v4 seq 5 permit 10.10.13.0/24
ip prefix-list SW1v4 seq 10 permit 10.10.15.0/24
ip prefix-list SW1v4 seq 15 permit 10.10.23.0/24
ip prefix-list SW1v4 seq 20 permit 10.10.25.0/24
ip prefix-list SW2v4 seq 5 permit 10.10.12.0/24
ip prefix-list SW2v4 seq 10 permit 10.10.22.0/24
ip prefix-list SW2v4 seq 15 permit 10.10.120.0/24
```

```
ip prefix-list SW2v4 seq 20 permit 10.10.150.0/24
!
ipv6 prefix-list SW1v6 seq 5 permit 2001:10:1:13::/64
ipv6 prefix-list SW1v6 seq 10 permit 2001:10:1:23::/64
ipv6 prefix-list SW1v6 seq 15 permit 2001:10:1:15::/64
ipv6 prefix-list SW1v6 seq 20 permit 2001:10:1:25::/64
ipv6 prefix-list SW2v6 seq 5 permit 2001:10:1:12::/64
ipv6 prefix-list SW2v6 seq 10 permit 2001:10:1:22::/64
ipv6 prefix-list SW2v6 seq 15 permit 2001:10:1:120::/64
ipv6 prefix-list SW2v6 seq 20 permit 2001:10:1:150::/64
!
route-map SW2v4 permit 10
 match ip address prefix-list SW2v4
 set as-path prepend 65000
!
route-map SW2v4 permit 20
!
route-map SW1v4 permit 10
 match ip address prefix-list SW1v4
 set as-path prepend 65000
!
```

```
route-map SW1v4 permit 20
!
route-map SW2v6 permit 10
 match ipv6 address prefix-list SW2v6
 set as-path prepend 65000
!
route-map SW2v6 permit 20
!
route-map SW1v6 permit 10
 match ipv6 address prefix-list SW1v6
 set as-path prepend 65000
!
route-map SW1v6 permit 20
!
router bgp 65003
 neighbor 10.10.1.1 route-map SW1v4 in
 neighbor 10.10.10.1 route-map SW2v4 in
 address-family ipv6 unicast
 neighbor 2001:10:1:1::1 route-map SW1v6 in
 neighbor 2001:10:1:2::1 route-map SW2v6 in
 exit-address-family
!
```

配置验证：特权模式

```
show bgp ipv6     //（交换机）查看IPv6 bgp路由表
show ip bgp      //(交换机、路由器)查看IPv4 bgp路由器
```

10.利用 BGP MPLS VPN 技术，RT1 与 RT2 以太链路间运行多协议标签交换、标签分发协议。RT1与RT2间创建财务VPN实例，名称为Finance，RT1 的 RD 值为 1:1，export rt 值为 1:2，import rt 值为 2:1；RT2的 RD 值为 2:2。通过两端 loopback1 建立 VPN 邻居，分别实现两端loopback5 ipv4 互通和 ipv6 互通。

RT1：全局配置模式

```
ipv6 vrf Finance
 rd 1:1
 route-target import 2:1
 route-target export 1:2
!
mpls ip
mpls ldp router-id 10.10.5.1
!
ip vrf Finance
```

```
 rd 1:1
 route-target export 1:2
 route-target import 2:1
!
interface Loopback5
 ip vrf forwarding Finance
 ip address 10.10.5.5 255.255.255.255
 ipv6 enable
 ipv6 address 2001:10:1:5::5/128
 ipv6 vrf forwarding Finance
!
interface GigaEthernet0/0
 mpls ip
 mpls ldp enable
!
router bgp 65001
 address-family vpnv4
 neighbor 10.10.6.1 activate
exit-address-family
 address-family vpnv6
 neighbor 10.10.6.1 activate
exit-address-family
 address-family ipv4 vrf Finance
 network 10.10.5.5/32
 exit-address-family
 address-family ipv6 vrf Finance
 network 2001:10:1:5::5/128
 exit-address-family
!
```

```
RT1#show ip bgp vpnv4 all summary
router bgp 65001
  BGP local AS: 65001, router identifier: 10.10.5.1
  IGP synchronization check: disabled
  Default local preference: 100
  Distance: external 20 internal 200 local 200
  Timers: keepalive 30, holdtime 90
  3 network entries using 600 bytes of memory
  3 path entries using 240 bytes of memory
  22 BGP path attribute entries using 4400 bytes of memory
  3 BGP AS-PATH entries using 72 bytes of memory
  2 BGP extended community entries using 40 bytes of memory

Neighbor        Ver         AS  State/Pref  Up/Down  FD  OutQ MsgRcvd MsgSent   VRF
10.10.6.1       4        65002           1 01:11:41    41    0     149     151

  Number of configured neighbors: 1

RT1#show ip bgp vpnv4 vrf Finance
BGP table version is 0, local router ID is 10.10.5.1
Status codes: s suppressed, d damped, h history, * valid, > best,
i - internal, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path
Route Distinguisher: 1:1 (default for vrf Finance, vrf id: 1)
*> 10.10.5.5/32     0.0.0.0                  0         32768 i
*> 10.10.6.5/32     10.10.6.1                0             0 65002 i
```

```
RT1#show ip bgp vpnv6 vrf Finance
BGP table version is 0, local router ID is 10.10.5.1
Status codes: s suppressed, d damped, h history, * valid, > best,
i - internal, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path
Route Distinguisher: 1:1 (default for vrf Finance, vrf id: 1)
*> 2001:10:1:5::5/128
                    ::                       0         32768 i
*> 2001:10:1:6::5/128
                    ::ffff:10.10.6.1
                                             0             0 65002 i
RT1#show ip bgp ipv6 unicast summary
router bgp 65001
  BGP local AS: 65001, router identifier: 10.10.5.1
  IGP synchronization check: disabled
  Default local preference: 100
  Default maximum routes limit: 1280000
  Distance: external 20 internal 200 local 200
  Timers: keepalive 30, holdtime 90
  21 network entries using 4200 bytes of memory
  21 path entries using 1680 bytes of memory
  22 BGP path attribute entries using 4400 bytes of memory
  3 BGP AS-PATH entries using 72 bytes of memory
  2 BGP extended community entries using 40 bytes of memory

Neighbor        Ver         AS  State/Pref  Up/Down  FD  OutQ MsgRcvd MsgSent   VRF
2001:10:1:1::1  4        65001           4 01:12:59    38    0     174     151
2001:10:1:2::1  4        65001           5 01:13:10    36    0     175     152
2001:10:1:6::1  4        65002           1 01:12:21    42    0     149     149
```

```
RT1#ping6 -vrf Finance -s 2001:10:1:5::5 2001:10:1:6::5
PING 2001:10:1:6::5 (2001:10:1:6::5): 56 data bytes
!!!!!
--- 2001:10:1:6::5 ping6 statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 0/0/0 ms
RT1#ping -vrf Finance -i 10.10.5.5 10.10.6.5
PING 10.10.6.5 (10.10.6.5): 56 data bytes
!!!!!
--- 10.10.6.5 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 0/0/0 ms
```

RT2：全局配置模式

```
ipv6 vrf Finance
 rd 2:2
 route-target import 1:2
 route-target export 2:1
!
mpls ip
mpls ldp router-id 10.10.6.1
!
ip vrf Finance
 rd 2:2
 route-target export 2:1
 route-target import 1:2
!
interface Loopback5
 ip vrf forwarding Finance
 ip address 10.10.6.5 255.255.255.255
 ipv6 enable
 ipv6 address 2001:10:1:6::5/128
 ipv6 vrf forwarding Finance
!
```

```
interface GigaEthernet0/0
 mpls ip
 mpls ldp enable
!
router bgp 65002
 address-family vpnv4
 neighbor 10.10.5.1 activate
exit-address-family
 address-family vpnv6
 neighbor 10.10.5.1 activate
exit-address-family
 address-family ipv4 vrf Finance
 network 10.10.6.5/32
 exit-address-family
 address-family ipv6 vrf Finance
 network 2001:10:1:6::5/128
 exit-address-family
!
```

```
RT2#show ip bgp vpnv4 all summary
router bgp 65002
  BGP local AS: 65002, router identifier: 10.10.6.1
  IGP synchronization check: disabled
  Default local preference: 100
  Distance: external 20 internal 200 local 200
  Timers: keepalive 30, holdtime 90
  3 network entries using 600 bytes of memory
  3 path entries using 240 bytes of memory
  14 BGP path attribute entries using 2800 bytes of memory
  3 BGP AS-PATH entries using 72 bytes of memory
  2 BGP extended community entries using 40 bytes of memory

Neighbor        Ver         AS  State/Pref  Up/Down  FD  OutQ MsgRcvd MsgSent   VRF
10.10.5.1       4        65001           1 01:18:22    43    0     163     164

  Number of configured neighbors: 1

RT2#show ip bgp vpnv4 vrf Finance
BGP table version is 0, local router ID is 10.10.6.1
Status codes: s suppressed, d damped, h history, * valid, > best,
i - internal, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path
Route Distinguisher: 2:2 (default for vrf Finance, vrf id: 1)
*> 10.10.5.5/32     10.10.5.1                0             0 65001 i
*> 10.10.6.5/32     0.0.0.0                  0         32768 i
```

```
RT2#show ip bgp vpnv6 vrf Finance
BGP table version is 0, local router ID is 10.10.6.1
Status codes: s suppressed, d damped, h history, * valid, > best,
i - internal, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path
Route Distinguisher: 2:2 (default for vrf Finance, vrf id: 1)
*> 2001:10:1:5::5/128
                    ::ffff:10.10.5.1
                                             0             0 65001 i
*> 2001:10:1:6::5/128
                    ::                       0         32768 i
RT2#show ip bgp ipv6 unicast summary
router bgp 65002
  BGP local AS: 65002, router identifier: 10.10.6.1
  IGP synchronization check: disabled
  Default local preference: 100
  Default maximum routes limit: 1280000
  Distance: external 20 internal 200 local 200
  Timers: keepalive 30, holdtime 90
  21 network entries using 4200 bytes of memory
  21 path entries using 1680 bytes of memory
  14 BGP path attribute entries using 2800 bytes of memory
  3 BGP AS-PATH entries using 72 bytes of memory
  2 BGP extended community entries using 40 bytes of memory

Neighbor        Ver         AS  State/Pref  Up/Down  FD  OutQ MsgRcvd MsgSent   VRF
2001:10:1:5::1  4        65001           9 01:19:36    44    0     162     165

  Number of configured neighbors: 1
```

```
RT2#ping6 -vrf Finance -s 2001:10:1:6::5 2001:10:1:5::5
PING 2001:10:1:5::5 (2001:10:1:5::5): 56 data bytes
!!!!!
--- 2001:10:1:5::5 ping6 statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 0/0/0 ms
RT2#ping -vrf Finance -i 10.10.6.5 10.10.5.5
PING 10.10.5.5 (10.10.5.5): 56 data bytes
!!!!!
--- 10.10.5.5 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 0/0/0 ms
```

配置验证：特权模式

```
show ip bgp vpnv4 all summary        //查看BGP vpnv4邻居
show ip bgp vpnv4 vrf Finance        //查看BGP vpnv4路由信息
show ip bgp vpnv6 vrf Finance
show ip bgp ipv6 unicast summary
```